Task 5 Predictive Modelling Report

Gradient boosting models were trained to predict air temperature and PM10 concentration from the merged hourly weather and PM10 dataset.

Dataset Summary

- Source file: task5_weather_pm10_hourly_master.csv
- Dataset rows: 24,096
- Dataset range: 2023-03-17 11:00:00 to 2026-05-01 00:00:00
- Training period: 2023-03-18 11:00:00 to 2026-04-15 23:00:00
- Testing period: 2026-04-16 to 2026-05-01
- Testing rows: 361

Model Metrics

Target	Train rows	Test rows	MAE	MSE	RMSE	R2
temperature	23,711	361	0.368	0.266	0.515	0.987
pm10	23,711	361	1.529	5.286	2.299	0.776

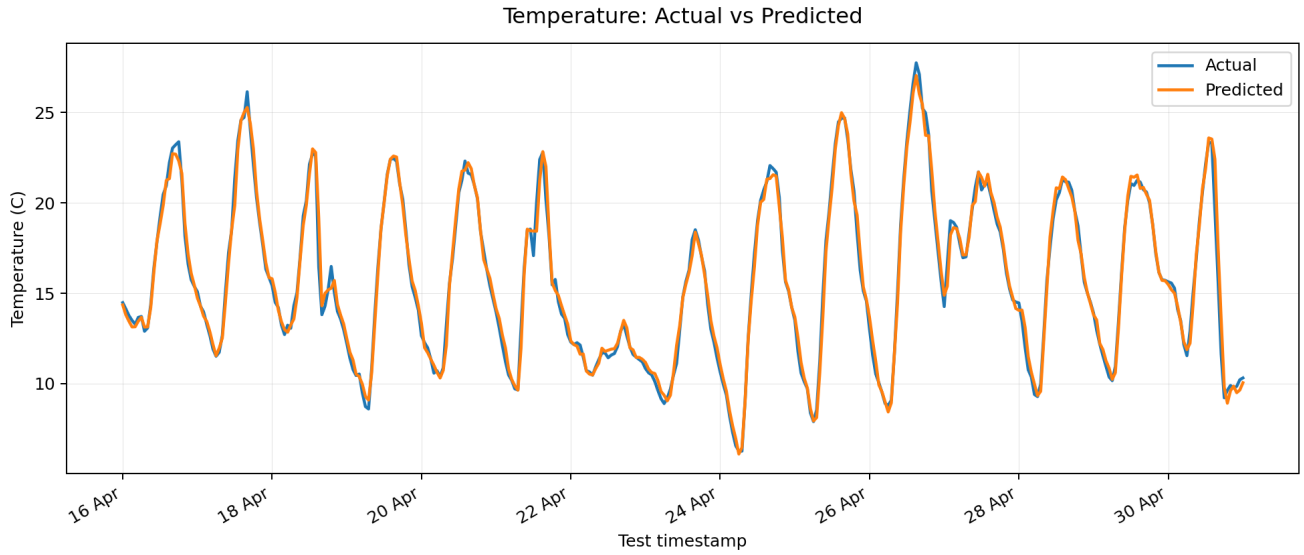
Methodology

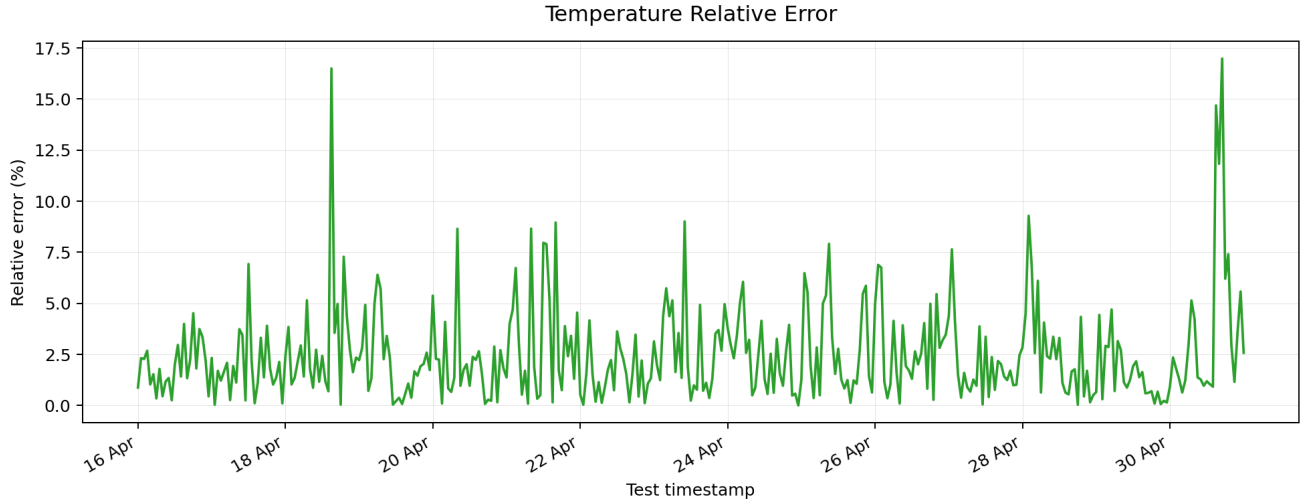
- The merged hourly CSV was loaded and sorted by timestamp.
- The latest 15 days were held out as the testing set; earlier rows were used for training.
- Separate gradient boosting regression models were trained for air temperature and PM10.
- Each model used all available feature columns except its own current target column.
- Feature engineering included time features, cyclical time encodings, wind-direction sine/cosine values, one-hour lags, 24-hour lags, and rolling averages.
- The model used HistGradientBoostingRegressor with max_iter=500, learning_rate=0.04, l2_regularization=0.05, and random_state=42.

Metric Definitions

- MSE is the average squared prediction error. Lower values indicate smaller errors, but the unit is squared.
- RMSE is the square root of MSE. It is easier to interpret because it is in the same unit as the target variable.
- R2 measures how much variation in the target is explained by the model. A value closer to 1 indicates a stronger fit.

Temperature Prediction





PM10 Prediction

